Calculating M-scheme matrix elements from coupled two-body RMEs

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(Dated: August 28, 2020)

Numerous conventions in the literature exist for the definition of the reduced matrix element in the Wigner-Eckart theorem. Here we derive the relationship between matrix elements (of a spherical tensor operator) in the uncoupled basis to reduced matrix elements in the coupled basis. We adopt the following convention for the Clebsch-Gordan coefficients and Wigner 3J symbols [1]

$$(j_a m_a j_b m_b|JM) = (-1)^{j_b-j_a-M}\,\hat{J}\begin{pmatrix} j_a & j_b & J \\ m_a & m_b & -M \end{pmatrix} \tag{1}$$

where $\hat{J} = \sqrt{2J+1}$.

First, we relate the coupled and uncoupled antisymmetrized two-particle bases:

$$\begin{aligned} |am_a bm_b\rangle_{\text{AS}} &= \sum_J (j_a m_a j_b m_b|JM)\,|ab;JM\rangle_{\text{AS}} \qquad (2)\\ &= \sum_J (-1)^{j_b-j_a-M}\,\hat{J}\begin{pmatrix} j_a & j_b & J \\ m_a & m_b & -M \end{pmatrix}|ab;JM\rangle_{\text{AS}} \qquad (3) \end{aligned}$$

Next we insert these into the matrix element:

$$\begin{aligned} &\langle am_a bm_b|\,\mathcal{O}\,|cm_c dm_d\rangle \\ &= \left(\sum_{J'} (-1)^{j_b-j_a-M'}\,\hat{J}'\begin{pmatrix} j_a & j_b & J' \\ m_a & m_b & -M' \end{pmatrix}\langle ab;J'M'|_{\text{AS}}\right)\mathcal{O} \\ &\quad\times\left(\sum_J (-1)^{j_d-j_c-M}\,\hat{J}\begin{pmatrix} j_c & j_d & J \\ m_c & m_d & -M \end{pmatrix}|cd;JM\rangle_{\text{AS}}\right) \\ &= (-1)^{j_b-j_a}(-1)^{j_d-j_c}(-1)^{-M'-M} \\ &\quad\times\sum_{J',J}\hat{J}'\hat{J}\begin{pmatrix} j_a & j_b & J' \\ m_a & m_b & -M' \end{pmatrix}\begin{pmatrix} j_c & j_d & J \\ m_c & m_d & -M \end{pmatrix}\langle ab;J'M'|\,\mathcal{O}\,|cd;JM\rangle_{\text{AS}} \end{aligned}$$

We use the Wigner-Eckart theorem to write the matrix element in terms of reduced matrix elements

$$\langle \alpha J' M' | \, \mathcal{O}_{J_0 M_0} \, | \beta J M \rangle = (-1)^{J'-M'} \begin{pmatrix} J' & J_0 & J \\ -M' & M_0 & M \end{pmatrix} \langle \alpha J' || \mathcal{O}_{J_0} || \beta J \rangle$$

Finally, we express the uncoupled matrix element in terms of the coupled reduced matrix element:

$$\begin{aligned}
&\langle a m_a b m_b | \, \mathcal{O}_{J_0 M_0} \, | c m_c d m_d \rangle \\
&= (-1)^{j_b - j_a} (-1)^{j_d - j_c} (-1)^{-M'-M} \\
&\qquad \times \sum_{J',J} \hat{J}' \hat{J} \begin{pmatrix} j_a & j_b & J' \\ m_a & m_b & -M' \end{pmatrix} \begin{pmatrix} j_c & j_d & J \\ m_c & m_d & -M \end{pmatrix} \langle ab; J' M' | \, \mathcal{O} \, | cd; JM \rangle_{\mathrm{AS}} \\
&= (-1)^{j_b - j_a} (-1)^{j_d - j_c} (-1)^{-M'-M} \\
&\qquad \times \sum_{J',J} (-1)^{J'-M'} \hat{J}' \hat{J} \begin{pmatrix} j_a & j_b & J' \\ m_a & m_b & -M' \end{pmatrix} \begin{pmatrix} j_c & j_d & J \\ m_c & m_d & -M \end{pmatrix} \begin{pmatrix} J' & J_0 & J \\ -M' & M_0 & M \end{pmatrix} \langle ab; J' || \mathcal{O}_{J_0} || cd; J \rangle
\end{aligned}$$

If we assume that $j_a, j_b, j_c, j_d \in \mathbb{Z} + \frac{1}{2}$ and that $J', M', J, M, J_0, M_0 \in \mathbb{Z}$, we can simplify and rearrange the phase factors to a form which can be implemented in `mfdn-transitions`:

$$\begin{aligned}
&\langle a m_a b m_b | \, \mathcal{O}_{J_0 M_0} \, | c m_c d m_d \rangle = \\
&(-1)^{j_b - j_a + j_d - j_c - M} \sum_{J'} (-1)^{J'} \hat{J}' \begin{pmatrix} j_a & j_b & J' \\ m_a & m_b & -M' \end{pmatrix} \sum_{J} \hat{J} \begin{pmatrix} j_c & j_d & J \\ m_c & m_d & -M \end{pmatrix} \begin{pmatrix} J' & J_0 & J \\ -M' & M_0 & M \end{pmatrix} \langle ab; J' || \mathcal{O}_{J_0} || cd; J \rangle
\end{aligned} \tag{4}$$

[1] J. Suhonen, *From Nucleons to Nucleus* (Springer-Verlag, Berlin, 2007).